Guiding Evolutionary Algorithms with Large Language Models to Learn Fuzzy Cognitive Maps

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Abstract

Fuzzy Cognitive Maps (FCMs) are interpretable simulation models that represent causal relationships between concepts as a weighted digraph with labeled nodes. They are commonly used to examine the structure (e.g., what concepts are critical to spreading the effects of an intervention?) and the long-term behavior of a system (e.g., if we increase fruit availability, how will it affect fruit consumption?). Modelers build FCMs by leveraging participants' knowledge or machine learning and data on the system. Finding enough knowledgeable participants to construct an FCM can be challenging, and learning algorithms often produce overly dense FCMs, prompting the development of methods to learn sparse FCMs that more closely resemble participant-built FCMs. However, the learned edges do not correspond to causal relationships as in participant-built FCMs, making structural analysis and interpretation difficult or impossible, which are two primary motivations for using FCMs. Thus, we use large language models (LLMs) to identify edges corresponding to causal relationships and their direction of causality among a list of concepts. We guide the generation of FCMs using these relationships as constraints in CMA-ES, a ubiquitous, state-of-the-art evolutionary algorithm. We apply our method to three real-world case studies. Across case studies, LLMs, and CMA-ES parameters, our method successfully learns sparse FCMs that fit the data well with edges that represent causal relationships, even when there is no corresponding participant-built FCM. Moreover, the learned FCMs are sparser than the corresponding participant-built ones, demonstrating our method selectively includes causal relationships, which is essential in building a trustworthy and interpretable FCM.

Keywords: Fuzzy Cognitive Maps, Large Language Models, Evolutionary Algorithms, CMA-ES

1 Introduction

Fuzzy Cognitive Maps (FCMs) are semi-quantitative simulation models that represent causal relationships between concepts in a system of interest as a weighted digraph with labeled nodes [1]. Thus, they are easy to interpret and visualize, which explains their frequent use in participatory modeling projects that include community members and subject-matter experts rather than modelers [2–4]. FCMs are commonly used to examine the *structure* (e.g., what are the most important concepts? [5, 6]) and the long-term *behavior* of a system (e.g., if we increase the availability of healthy foods in a community, how will that affect the prevalence of diabetes?), supporting the development of targeted interventions. Modelers have applied FCMs to numerous diverse domains, such as medicine [7, 8], education [9, 10], environmental management [11, 12], and engineering [10, 13].

FCMs are built by leveraging stakeholders' knowledge or learning algorithms and real-world data. Modelers can quickly construct FCMs through facilitated interviews with participants lasting less than an hour [14] and aggregate individual participant-level FCMs into group-level models, enabling comparison of perspectives of distinct groups [15] (e.g., the traditional scientific community and Canadian Aboriginals [16]). Participants are often selective with the causal relationships they include in their model, producing sparse models [14]. However, finding knowledgeable participants for a given domain available to develop a model can be challenging, especially since several participants are often needed to produce a reliable group-level model where critical concepts and relationships are present [17]. Thus, numerous learning methods to automatically construct FCMs from data have been developed and extensively applied [18, 19], especially to data-rich domains such as engineering [13]. Yet, learning-based approaches often construct overly dense FCMs with edges that do not correspond to causal relationships in the data, making interpretation and structural analysis difficult or impossible [19, 20], which are two of the main reasons FCMs are chosen. If we learn a dense FCM, it improves accuracy but hinders interpretability, so it may more closely resemble an accurate but un-interpretable neural network than an interpretable, participant-built FCM. Thus, significant research has examined learning FCMs from data while retaining their interpretability [21, 22].

Researchers have proposed methods to learn sparse FCMs that more closely resemble those developed by participants and are easier to interpret. The sparse real-coded genetic algorithm employs a density parameter to guide the algorithm in generating an FCM with appropriate *density* (ratio of existing edges to all possible edges) [20]. They suggest a default value of 0.37 for the density parameter because they found the average density was 0.37 for 18 real-world, participant-built models [20], but the largest model they consider has 13 concepts, and we expect the density to decrease as the number of concepts increases [14]. Thus, the appropriate density parameter for a particular real-world case study is unknown, prompting the creation of a multiobjective evolutionary algorithm that generates numerous candidate FCMs that minimize *both* the error to the observed data and difference from the desired density, with each candidate FCM providing a different balance between sparsity and error [23]. In contrast, the quantum FCM uses a quantum evolutionary algorithm to determine the

structure (and thus the density) of the FCM and particle swarm optimization to specify the edge weights [24], and the iterative smoothing algorithm of structural sparsity uses elastic (combination of L_1 and L_2 regularization) and total variation regularization to learn a sparse FCM without overfitting the data or noise [25]. Although these approaches learn sparse FCMs more similar to participant-built FCMs, they do not leverage domain expertise and identify edges and their weights solely by optimizing a fit with the data, which can lead to edges that do not correspond to real-world causal relationships, have the wrong direction of causation, or with large departures in causal strength.

We use large language models (LLMs) to identify causal relationships present among input concepts and incorporate the identified relationships as constraints in evolutionary algorithms (EAs), specifically CMA-ES, to automatically learn sparse FCMs whose relationships correspond to causal relationships in the domain of interest, similar to participant-built FCMs. Our work investigates three research questions:

1. Can LLMs successfully guide EAs to learn FCMs from data? We propose and evaluate a method that fits synthetic data from participant-built FCMs and real-world longitudinal data.
2. Can the LLM-guided EA learn sparse FCMs?
3. What is the optimal configuration to guide FCM generation? There are several LLMs (each with a parameter) and parameters for EAs, so we analyze the effect of these settings across case studies.

The remainder of the paper is organized as follows. Section 2 introduces the fundamental concepts of FCMs, learning algorithms for FCMs, LLMs, and causality in LLMs. We specify our method for guiding FCM construction and apply it to three real-world case studies in section 3. Section 4 states our results, and section 5 discusses their importance and outlines directions for future work. We provide our code and results files at https://zenodo.org/records/11194343.

2 Background

2.1 Fuzzy Cognitive Maps

Fuzzy Cognitive Maps (FCMs) are weighted digraphs with labeled nodes [1], so they can easily be visualized (Figure 1), making them popular choices for participatory modeling contexts when additional stakeholders other than modelers are involved such as community members affected by the model's decisions [2–4]. Moreover, they are easy to interpret. Edge weights are restricted to the interval [-1, 1], where -1 is the strongest possible negative causal relationship (an increase in the source leads to a decrease in the target), 1 is the strongest possible positive causal relationship (an increase in the source leads to an increase in the target), and 0 means there is no causal relationship. Thus, modelers can compare the relative strength of causal relationships. For example, the edge from body weight to diabetes has a weight of 1, and the edge from smoking to diabetes has a weight of 0.04, so the impact of body weight on diabetes is much greater than that of smoking (Figure 1). Additionally, FCMs assign each concept a

value in the interval [0, 1] to enable simulation, which permits investigating the long-term behavior of the system through *what-if* questions (e.g., what if we reduce the availability of alcohol? How will it affect the prevalence of diabetes?). A value of 0 means there is as little of the concept as possible (e.g., no diabetes), whereas a value of 1 means there is as much as possible (e.g., everyone has diabetes).

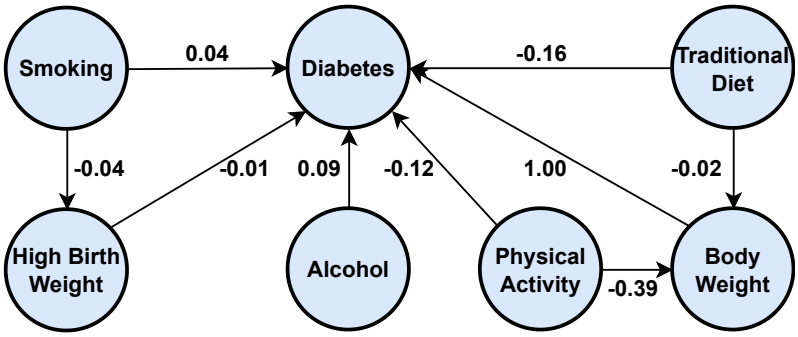

Fig. 1 A subset of the scientific knowledge FCM on diabetes from [16].

Formally, an FCM is represented as a tuple (A^t, W, f), where A^t is the state vector of M concept values at the t-th iteration, W is the weight matrix containing the edge weights for each causal relationship, and f is a threshold function used to limit concept values to the interval [0, 1] or [-1, 1]. Popular choices for f are the sigmoid function $f(x) = \frac{1}{1+e^{-\lambda x}}$, where λ controls the steepness of the function, and the hyperbolic tangent function $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, which restrict concepts values to [0, 1] and [-1, 1], respectively [19]. FCMs perform simulation in discrete iterations with synchronous updates (i.e., all concept values are stored in a buffer and updated simultaneously). Simulation continues until a set of output concepts (e.g., diabetes) changes by less than a small, user-defined value or a maximum number of iterations is reached. There are two commonly used equations for FCM equations. The FCM 'without memory' calculates the value for the concept A_j at an iteration $t+1$ as follows:

$$A_j^{t+1} = f\underbrace{\left(\sum_{\substack{i=1 \\ i \neq j}}^{M} W_{ij} \times A_i^t\right)}_{\text{total input to node}}, \tag{1}$$

where W_{ij} is the edge weight for the edge from the concept A_i to A_j and A_i^t is the value for the concept A_i at iteration t. The condition $i \neq j$ prevents the previous value of the node from affecting its current value. Thus, modelers proposed an alternative equation 'with memory' that allows a concept's previous value to affect its current

one [26]:

$$A_j^{t+1} = f\left(\underbrace{\left(\sum_{\substack{i=1\\i\neq j}}^{M} W_{ij} \times A_i^t\right)}_{\text{total input to node}} + \underbrace{A_j^t}_{\text{memory}}\right) \tag{2}$$

2.2 Learning Algorithms for Fuzzy Cognitive Maps

2.2.1 Overview of Approaches

Learning algorithms for FCMs determine the weight matrix W given data, expert input, or both [19]. There are three main categories of learning algorithms for FCMs: Hebbian-based, error-driven, and hybrid approaches. First, Hebbian-based learning algorithms are often unsupervised (i.e., they do not require input/output pairs) and modify an initial FCM built by participants [19]. Differential Hebbian Learning (DHL) was the first Hebbian-based learning algorithm for FCMs. It updates an edge weight W_{ij} by a constant factor if both the source concept A_i and target concept A_j change their values in the same iteration [27]. Balanced Differential Algorithm builds on DHL by considering all concept changes that occur in the same iteration with the same directionality (i.e., increase or decrease), capturing more of the system behavior [28]. In Active Hebbian Learning (AHL), concept updates are asynchronous, and participants specify a sequence of concept updates to guide the asynchronous updates [29]. Participants also provide the initial structure of the FCM (i.e., what weight matrix entries are nonzero); however, the algorithm can update all weights other than self-loops corresponding to 'memory' in the FCM, potentially distorting the initial expert-provided structure by adding or removing edges or changing the edge weights directionality [29]. To address this limitation, Nonlinear Hebbian Learning (NHL) elicits the initial weights and their direction and plausible ranges for concept values from participants [30], but, unlike AHL, NHL retains the structure provided by the participants, ensuring the structure is interpretable and relationships correspond to causal relationships. However, the dependence on participant input and the inability to leverage data on the system are limitations for both AHL and NHL.

Second, error-driven learning methods minimize an error function that depends on the distance from the learned model to observed data (e.g., the difference between observed and learned concept values at each iteration) [19]. They require defining an error function and at least one input/output state pair (i.e., initial state vector and corresponding final concept values) and often use computationally demanding optimization techniques such as EAs [31–33]. For example, an existing method minimizes the absolute difference between the learned and observed concept values over the M concepts and K input/output state pairs:

$$E(W) = \sum_{k=1}^{K}\sum_{i=1}^{M} |A_{ki} - \tilde{A_{ki}}|, \tag{3}$$

where W is the weight matrix, $E(W)$ is the error of the weight matrix, A_{ki} is the learned value and $\tilde{A_{ki}}$ is the observed value [34]. Some methods require multiple input/output pairs [34, 35], whereas others learn FCMs from one input/output pair [31–33, 36]. Error-driven approaches do not require participant input, but they often learn dense models whose edges do not correspond to causal relationships in the system of interest, making them difficult or impossible to interpret and impairing structural analsyis [19, 37]. Moreover, as the number of concepts in the FCM grows, computational overhead can be prohibitive. Thus, modelers have developed learning algorithms for large maps up to 100 [38] or 300 concepts [39].

Hybrid learning approaches combine Hebbian-based, usually NHL, and error-driven learning, often refining an initial weight matrix provided by participants given data on the system [19]. Of the three categories, hybrid learning has the fewest approaches [19]. Researchers have coupled differential evolution and NHL, finding the coupling can learn appropriate weight matrices that lead the FCM to desired states in real-world problems [40]. Combining real-coded genetic algorithms and NHL effectively integrated participant knowledge with real-world data to solve a partner selection problem [41]. Additionally, modelers trained an FCM with NHL to learn weights close to an optimal but unknown structure before applying the extended great deluge algorithm to optimize the structure through error minimization [42].

2.2.2 Evolutionary Algorithms

Evolutionary algorithms (EAs) are a popular type of search algorithms often used to optimize parameters [43, 44], including edge weights in FCMs [33, 45, 46]. Genetic algorithms and evolution strategies are two popular classes of EAs. EAs navigate the search space using a fitness function, with a higher fitness corresponding to a better fit, similar to how error-driven learning approaches use an error function with a lower error indicating a superior fit. EAs have three main steps. First, an initial candidate population is randomly generated, with each individual often called a *genome* or chromosome and each entry in the genome referred to as a *gene*. Second, the EA selects genomes to retain using the fitness function. Third, the chosen genomes produce new ones via mutation (randomly changing genes to add diversity to the population), crossover (selecting individual genes from the parent genomes), recombination (e.g., taking the mean of the parent's genes for a gene), or a combination of approaches. Every iteration of these steps is called a *generation*, and iteration continues until a maximum number of generations is reached or the fitness function does not improve.

By default, EAs do not handle constraints. However, modelers can incorporate constraints into EAs through the use of a penalty function, which assigns a lower fitness to invalid genomes (e.g., candidate FCM weight matrices with entries outside of the interval [-1, 1]) [47, 48]. Thus, over generations, genomes that violate the constraints will be removed from the population. For example, modelers could assign a constant penalty to invalid genomes or a penalty that depends on the distance to the closest valid genome, such as the quadratic distance $(x - x_0)^2$, where x_0 is the center of the valid zone (Figure 2).

EAs for FCMs are usually evaluated by the error/fitness to the real-world data [20, 23, 24, 45, 46] and compared to the participant-built FCM if one exists [20, 23, 24].

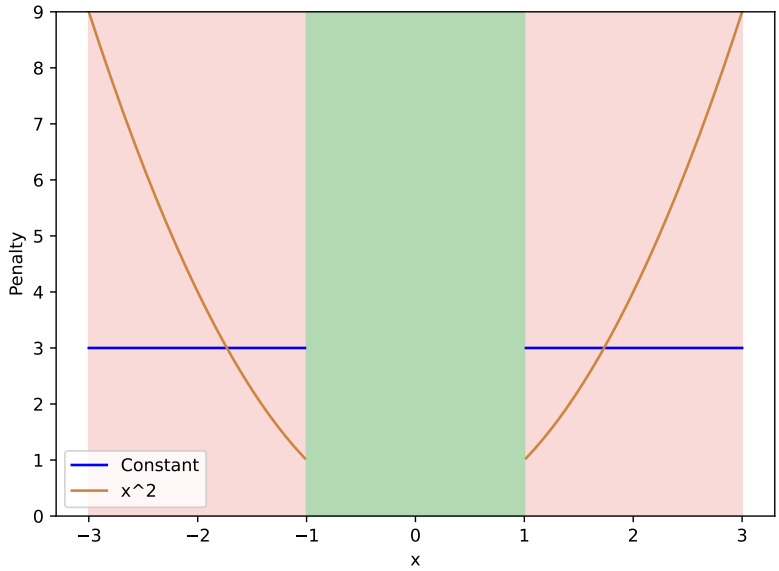

Fig. 2 Example penalties for genomes outside the valid range of [-1, 1].

While EAs have been widely applied to FCMs, they often learn overly dense FCMs with edge weights that do not correspond to causal relationships. Thus, variants of EAs to learn sparse FCMs have been proposed, such as the sparse real-coded genetic algorithm [20], multiobjective evolutionary algorithm [23], and quantum FCM that uses a quantum evolutionary algorithm [24]. However, these approaches do not leverage domain knowledge to identify edges. Consequently, there is a risk of creating edges that do not represent real-world causal relationships, have the incorrect direction of causality, or have significant discrepancies in causal strength.

Covariance matrix adaptation evolution strategy (CMA-ES) is a state-of-the-art evolution strategy for non-linear, non-convex black-box optimization. It is efficient for large search spaces [49], and modelers have applied it to learn FCMs [45]. CMA-ES generates candidate genomes from an initial multivariate normal distribution, and modelers must provide appropriate initial centroid values for the multivariate normal distribution and an initial standard deviation for the distribution given their use case. Additional parameters, such as the population size, have default values provided and may be changed by the user, although the selection of these optional parameters is relatively uncritical and will not hinder the algorithm. CMA-ES evaluates the genomes according to the fitness function and produces new genomes through weighted recombination. Additionally, it automatically adapts the step size and covariance matrix to efficiently find optimal solutions. Similar to EAs in general, researchers have used penalty functions to encode constraints in CMA-ES [50].

2.3 Large Language Models

Giabbanelli will do

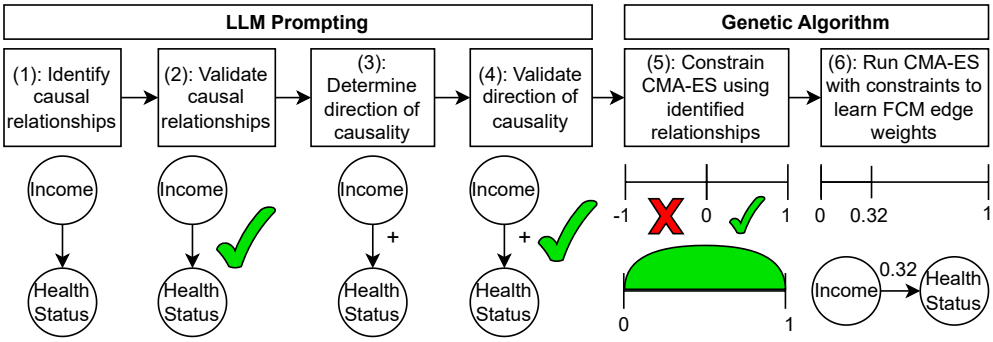

Fig. 3 Six-step process to synthesize LLMs and CMA-ES to learn FCMs

2.4 Causality in Large Language Models

Giabbanelli will do. Compare CoVe to other methods for validation. Search for papers that validate graphical models / knowledge graphs, particularly as two steps process (include edge yes/no, valid label yes/no). Also, here is the paper on FCMs for causal inference: https://ieeexplore.ieee.org/abstract/document/10495183. Was not sure how to tie that in, other than maybe in the discussion.

3 Methods

3.1 Algorithm for Guiding Fuzzy Cognitive Map Construction

We propose a six-step procedure to guide the construction of FCMs leveraging the combination of LLMs and CMA-ES (Figure 3). First, we prompt the LLM to list all causal relationships between the concepts in the system.

> List all causal relationships between a list of concepts with each relationship as a pair, ensuring each concept is involved in at least one relationship. For example, given [‘coffee consumption’, ‘sleep’, ‘energy level’], return [(‘coffee consumption’, ‘sleep’), (‘sleep’, ’energy level’), (‘coffee consumption’, ‘energy level’)] with no other text.
>
> `[concepts go here]`

We ask the LLM to include each concept in at least one relationship because the concept will not affect FCM simulation if it is not in a relationship (i.e., it has no incoming or outgoing edges). Second, we leverage Chain-of-Verification (CoVe) [51] to confirm the causal relationships provided by the LLM in step one exist by asking a new instance of the LLM without access to the previous prompt and response if a causal relationship exists between the concepts for each identified relationship.

Return whether a causal relationship exists between the source and target concepts for each pair in a list. For example, given [('smoking', 'cancer'), ('ice cream sales', 'shark attacks')], return ['Y', 'N'] with no other text.
`[concepts go here]`

If the output is 'N' for a previously identified causal relationship, we eliminate it from the list of relationships, minimizing the inclusion of an incorrect edge. Third, we prompt the LLM to determine whether each causal relationship is positive or negative.

Given a list of causal relationships, return whether each is positive or negative. For example, given [('coffee consumption', 'sleep'), ('sleep', 'energy level'), ('coffee consumption', 'energy level')], return ['-', '+', '+'] with no other text.
`[causal relationships go here]`

Fourth, we again employ CoVe to verify the output by prompting a new instance of the LLM with the following prompt to confirm whether each relationship is positive or negative.

Return whether each causal relationship in a list is positive or negative. For example, given [('Average Fish Size', 'Commercial Fishing'), ('Commercial Fishing', 'Average Fish Size'),('Price', 'Commercial Fishing')], return ['+', '-', '+'] with no other text.
`[causal relationships go here]`

We eliminate relationships with disagreement between the two responses, ensuring the identified relationships and their directions have been validated by the LLM.

Fifth, we use the final list of causal relationships and their directions to constrain the edge weights for CMA-ES. We set all edge weights corresponding to causal relationships not identified by the LLM to zero and prevent CMA-ES from modifying them by omitting them from the genome. Additionally, for each relationship identified by the LLM, we constraint the corresponding edge weight to the interval [0, 1] if the relationship is positive and [-1, 0] if it is negative through the use of a penalty function, which is standard practice for incorporating constraints into EAs and CMA-ES [50, 52, 53]. Specifically, we reposition the genome to the closest valid region (e.g., -1 if one of the edge weights/genes was less than -1) and add a penalty based on the sum of squares error (SSE) from the genome to the closest valid genome:

$$\sum_{i=1}^{N}(I_i - V_i)^2, \tag{4}$$

where N is the number of genes/edge weights in the genome, I_i is the i-th gene for the invalid genome I, and V_i is the i-th gene for the closest valid genome V. Thus, invalid genomes are assigned a lower fitness based on how far they are from the closest valid genome, promoting selection of genomes that satisfy the encoded constraints.

Sixth, we run CMA-ES with the constraints encoded through the penalty function, an initial centroid centered on our search space (0.5 for positive relationships in [0, 1] and -0.5 for negative relationships in [-1, 0]), an initial standard deviation of $\sigma = 0.25$ to initially generate 95% of the data within the valid bounds (we have two standard deviations/0.5 from the initial centroid values to their bounds). We use the default values for the remaining parameters, including for population size with a value of $\lfloor 4 + 3 \times ln(n) \rfloor$ [49], where n is the number of free parameters (in our case, the number of identified relationships after step four).

3.2 Case Studies

We use GPT-4, GPT-4 Turbo, Claude 3 Opus, and Gemini 1.0 with five temperature settings (0, 0.25, 0.5, 0.75, and 1.00) to guide the construction of FCMs and evaluate the impact of LLM architecture and parameter settings on FCM construction. We use the latest stable (i.e., permanent, no longer updated) model version for each of the four LLMs available at the time of experimentation (end of April 2024) to increase the reproducibility of our work: gpt-4-0613, gpt-4-turbo-2024-04-09, claude-3-opus-20240229, and gemini-1.0-pro-001. Note that we use Gemini 1.0 instead of Gemini 1.5 because there is not a stable version of Gemini 1.5 at the time of experimentation.

We apply each model configuration to three real-world case studies created by different research groups: one on diabetes, municipal governance, and nutrition (Table 1). The diabetes and municipal governance case studies each have an FCM created by participants, which will serve as a ground-truth FCM, and the nutrition case study provides real-world longitudinal data without a ground-truth FCM. For the diabetes and municipal governance case studies with an FCM, we create a synthetic dataset to learn the FCM by performing simulation with the ground-truth FCM and recording the state values over iterations, whereas we directly apply the algorithm to the real-world longitudinal data for the nutrition case study (Figure 4). First, the diabetes case study creates two FCMs, one representing the knowledge of the Canadian aboriginal community and one the traditional scientific community's, on the determinants of diabetes among the Canadian aboriginal community [16]. We use the FCM built by the traditional scientific community. We input an initial state vector consisting of all 0.5s because the paper does not provide baseline values, and popular FCM software, such as Mental Modeler [54], uses 0.5 as a default value. Second, the municipal governance case study examines the critical elements of municipal governance influencing the short and long-term well-being of citizens in Ecuador, such as having a competent mayor [55]. To create the synthetic dataset, we use the baseline scenario values provided. Third, the nutrition case study investigates the effects of web-based, computer-tailored nutrition education interventions on self-reported fruit, vegetable, high-energy snack, and saturated fat intake [56]. We use a subset of the dataset focusing on self-reported fruit intake with 722 participants, each with three data points: one at baseline, after one month, and after four months.

Table 1 Characteristics of the three case studies.

Case study	# concepts	Ground-truth FCM	# edges	Density	Threshold Function	Memory	Ref
Diabetes	18	Yes	29	0.095	Sigmoid	False	[16]
Municipal governance	12	Yes	31	0.235	Sigmoid	True	[55]
Nutrition	16	No	N/A	N/A	Sigmoid	True	[56]

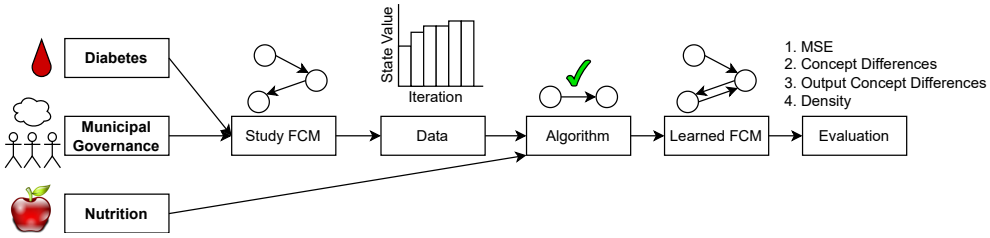

Fig. 4 Process for each case study.

3.3 Analysis

We perform enough replications of the proposed method to create an 85% confidence interval [57] on the final MSE obtained by CMA-ES for each model configuration and case study, with a bare minimum of five successful replications or 25 failed replications without a single success (e.g., an LLM may produce a difference number of positive and negative classifications in steps three and four). We evaluate each model configuration on the three case studies according to four metrics averaged across replications. First, we record the final mean MSE, with a lower value indicating a better fit to the data. Second, we measure the mean concept differences between the final learned and observed values for each concept averaged over concepts and replications. Third, we examine the mean difference for the output concept of interest. The output concept for the diabetes case study is diabetes, the municipal governance case study is the basic unsatisfied needs of the population in the canton, and the nutrition case study is fruit pieces per day. Finally, we evaluate the mean density to ensure our method learns sparse models. The density of an FCM is the fraction of all possible edges that exist:

$$D = \frac{|E|}{M(M-1)}, \tag{5}$$

where $|E|$ is the number of edges and M is the number of concepts.

Additionally, we conduct a sensitivity analysis of the population size, using the minimum suggested value of 5, n, and maximum recommended value of $2n+10$ [49] as alternate values. We use one-way ANOVA [58] to determine if the mean MSE differs across population sizes. We halt CMA-ES if the fitness measured by mean squared error (MSE) does not improve by more than 0.001 over 50 generations.

4 Results

Across case studies, LLM models, and temperatures, LLMs can successfully guide EAs to fit participant-built FCMs and real-world data without a corresponding FCM well, answering our first research question. The diabetes case study has the lowest MSE (Figure 5) and mean concept differences (Figure 6) across LLMs and temperatures. The municipal governance case study has the lowest difference for the output concept of interest for some configurations, but the diabetes case study has a lower difference for others (Figure 7). Despite lacking a ground-truth FCM, our method still successfully learns FCMs that fit the real-world nutrition case study.

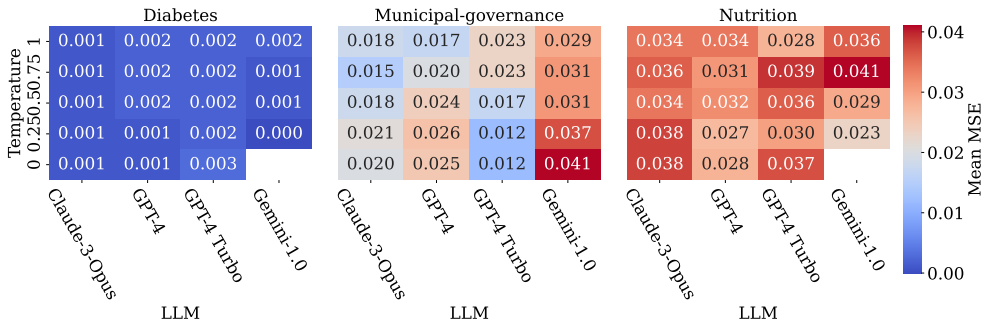

Fig. 5 Mean MSE for each model configuration and case study.

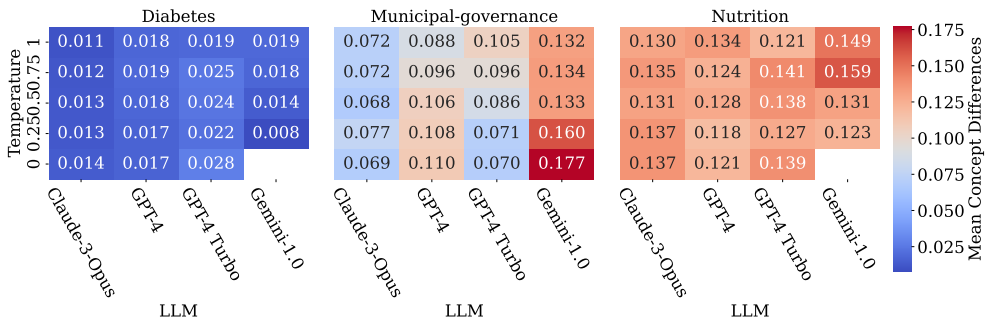

Fig. 6 Mean concept differences for each model configuration and case study.

Moreover, our method produces sparse FCMs (Figure 8), with almost every configuration producing FCMs with an average density of less than half of the participant-built FCM for the diabetes case study and a third for the municipal governance case study (Table 1), answering our second research question. Thus, our method is quite conservative when including causal relationships due to the inclusion of two

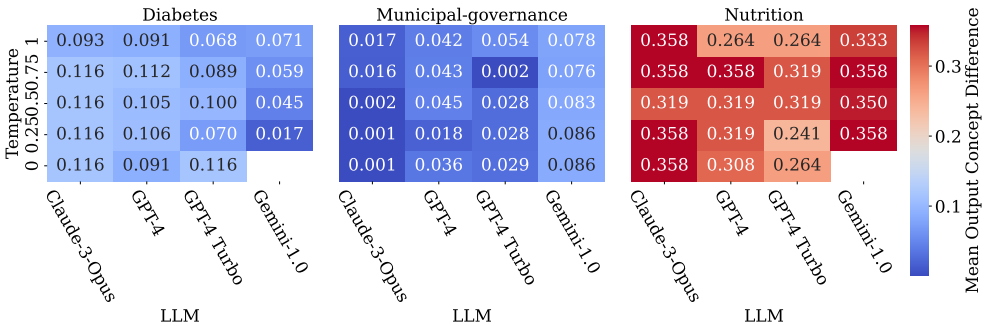

Fig. 7 Mean output concept difference for each model configuration and case study.

validation steps, one on the causal relationships and one on the direction of causality, minimizing false positives for causal relationships. For the nutrition case study, the models have a density of at most 0.075, well below the average participant-built FCM density of 0.37 [20]. While a low density is desirable, it must be balanced with identifying relationships that fit the system behavior. For example, Gemini 1.0 with temperature 0 on the municipal governance case study identifies one edge on average (Figure 9).

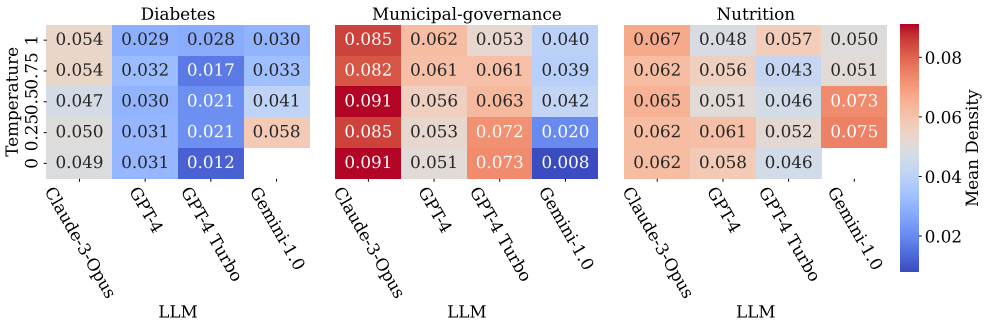

Fig. 8 Mean density for each model configuration and case study.

When investigating our third research question, we found the optimal model LLM and temperature depends on the desired tradeoff between density and error (denser FCMs tend to have smaller error [23]) and the case study. For example, Gemini 1.0 with temperature 0.25 has the lowest mean MSE, mean concept difference, and mean output concept difference but the highest density for the diabetes case study. On the other hand, for the municipal governance case study, Gemini 1.0 with temperature 0.25 has the second lowest density but the second highest mean MSE, mean concept difference, and is tied for the highest mean output concept difference. Temperature seems to have the smallest effect on Claude 3 Opus across case studies and metrics and the highest impact on Gemini 1.0. Gemini 1.0 with temperature 0 does not successfully identify

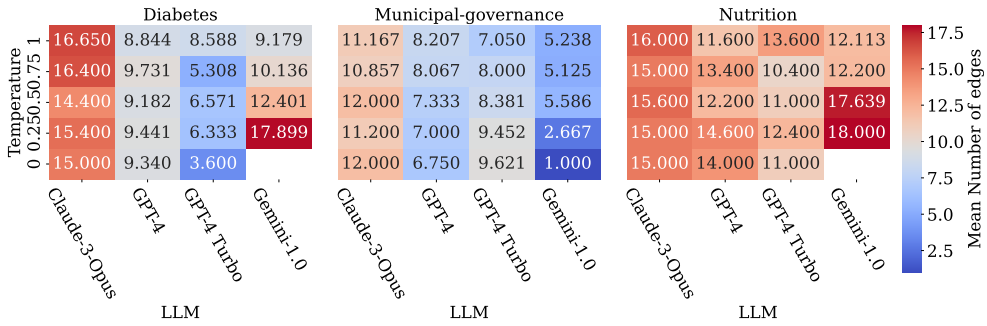

Fig. 9 Mean number of edges for each model configuration and case study.

and validate causal relationships for the nutrition and diabetes case studies, failing 25 runs with no successes (e.g., it produces a differing number of positive/negative classifications in steps three and four). Thus, if modelers do not want to optimize temperature, Claude 3 Opus may be the safest option for their case study, whereas Gemini 1.0 will likely need to be tuned and may produce better fitting or sparse models.

Population size significantly affects the mean MSE for Gemini 1.0 on the diabetes case study across temperatures and all but Gemini 1.0 with temperature 0.75 on the nutrition case study with a confidence level of 99% (Table 2). Population size does not affect any of the models for the municipal governance case study, demonstrating the effect of population size depends on the case study. However, when a significant difference exists, the default population size of $\lfloor 4 + 3 \times ln(n) \rfloor$ produces the lowest mean MSE, so the case study does not affect the optimal choice of population size.

5 Discussion

Numerous algorithms exist to learn FCMs from data without participant involvement, including those to learn sparse FCMs that more closely resemble participant-built FCMs. However, our approach is the first to use LLMs to automatically leverage domain expertise to identify causal relationships among the concepts in the data and restrict learning to these relationships. As a result of our first research question, we found LLMs can guide EAs in successfully learning FCMs that fit participant-built FCMs and real-world longitudinal data without a corresponding FCM well. Additionally, answering our second research question, the learned FCMs are sparse, and their edges correspond to causal relationships, supporting structural analysis and interpretation of FCMs as in participant-built FCMs. Moreover, the learned FCMs for the diabetes and municipal governance real-world case studies are sparser than their participant-built FCMs while exhibiting the same behavior, reflecting our method is selective in including causal relationships, which supports building a trustworthy and interpretable FCM. Thus, our method may help simplify FCMs while retaining their validity, an ongoing and impactful research endeavor [59]. Finally, for our third question of the optimal configuration, we found our method's performance depends on the

Table 2 Results of the sensitivity analysis, where the entries under diabetes, municipal governance, and nutrition are the p-values form the one-way ANOVA rounded to three decimal places. Note that N/A values occur for some configurations of Claude 3 Opus and Gemini 1.0 when all the MSE values are the same across replications and also for Gemini 1.0 with temperature 0 on the nutrition and diabetes case studies because they do not successfully return validated causal relationships.

LLM	Temperature	Diabetes	Municipal Governance	Nutrition
Claude 3 Opus	0	N/A	N/A	0.000
Claude 3 Opus	0.25	N/A	1.000	0.000
Claude 3 Opus	0.5	N/A	1.000	8.857×10^{-14}
Claude 3 Opus	0.75	N/A	1.000	2.534×10^{-9}
Claude 3 Opus	1.00	0.890	1.000	9.728×10^{-11}
GPT-4	0	0.992	0.996	1.188×10^{-7}
GPT-4	0.25	0.954	0.418	1.647×10^{-9}
GPT-4	0.5	0.719	0.985	5.296×10^{-9}
GPT-4	0.75	0.647	0.854	1.012×10^{-8}
GPT-4	1	0.771	0.821	0.001
GPT-4 Turbo	0	0.418	0.738	0.002
GPT-4 Turbo	0.25	0.521	0.717	8.533×10^{-5}
GPT-4 Turbo	0.5	0.321	0.874	5.959×10^{-11}
GPT-4 Turbo	0.75	0.948	1.000	0.001
GPT-4 Turbo	1	0.196	0.996	4.403×10^{-5}
Gemini 1.0	0	N/A	N/A	N/A
Gemini 1.0	0.25	0.000	0.464	7.955×10^{-8}
Gemini 1.0	0.5	5.809×10^{-89}	0.200	4.637×10^{-8}
Gemini 1.0	0.75	1.684×10^{-20}	0.539	0.068
Gemini 1.0	1	2.527×10^{-20}	0.582	3.263×10^{-6}

LLM, temperature, and case study, although the default value for population size in CMA-ES always produced the optimal results. As in prior work [23], we observed the tradeoff between density and error, with denser models tending to fit the data better. Thus, the optimal model configuration also depends on the user's preferences for reducing error at the expense of sparsity.

Our work has three main limitations. First, our method needs a list of concepts to identify causal relationships between them, but most learning algorithms assume the set of concepts is given [18, 19]. Future work could explore providing a few seed concepts or just the target domain and using the LLM to generate the remaining concept names when a list of names is unavailable. Second, our method and LLMs rely on temperature, which may need to be optimized. However, parameter optimization is inevitable in applied machine learning research where several algorithms (e.g., support vector machines, linear regression with elastic net regularization) and parameter values (e.g., different L_1 and L_2 coefficients in elastic net regularization) are applied. Moreover, Claude 3 Opus appears relatively unaffected by temperature on this task, so tuning may not be essential, especially if tuning is infeasible for a particular use case given time, computational, or financial constraints. Third, LLMs may lack knowledge of causal relationships for a domain. Thus, future work could use retrieval-augmented generation to provide additional external knowledge on the domain of interest to the LLM [60].

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Competing Interests

The authors did not receive support from any organization for the submitted work.

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Data availability

We include input files for the diabetes and municipal governance case studies and our results files at the following URL: https://zenodo.org/records/11194343. The data for the nutrition case study can be requested from the authors of the original study.

Materials availability

Not applicable

Code availability

We include our code at the following URL: https://zenodo.org/records/11194343.

Author contribution

Conceptualization: Philippe J. Giabbanelli; Data curation: Ryan Schuerkamp, Philippe J. Giabbanelli; Formal analysis: Ryan Schuerkamp; Funding acquisition: N/A; Investigation: Ryan Schuerkamp; Methodology: Ryan Schuerkamp, Philippe J. Giabbanelli; Project administration: Philippe J. Giabbanelli; Resources: Philippe J. Giabbanelli; Software: Ryan Schuerkamp; Supervision: Philippe J. Giabbanelli; Validation: Ryan Schuerkamp; Visualization: Ryan Schuerkamp; Writing – original draft: Ryan Schuerkamp, Philippe J. Giabbanelli.

References

[1] Kosko, B.: Fuzzy cognitive maps. International journal of man-machine studies **24**(1), 65–75 (1986)

[2] Voinov, A., Jenni, K., Gray, S., Kolagani, N., Glynn, P.D., Bommel, P., Prell, C., Zellner, M., Paolisso, M., Jordan, R., *et al.*: Tools and methods in participatory modeling: Selecting the right tool for the job. Environmental Modelling & Software **109**, 232–255 (2018)

[3] Özesmi, U., Özesmi, S.L.: Ecological models based on people's knowledge: a multi-step fuzzy cognitive mapping approach. Ecological modelling **176**(1-2), 43–64 (2004)

[4] Gray, S.A., Gray, S., De Kok, J.L., Helfgott, A.E., O'Dwyer, B., Jordan, R., Nyaki, A.: Using fuzzy cognitive mapping as a participatory approach to analyze change, preferred states, and perceived resilience of social-ecological systems. Ecology and Society **20**(2) (2015)

[5] Schuerkamp, R., Giabbanelli, P.J.: Analysis of fuzzy cognitive maps. In: Fuzzy Cognitive Maps: Best Practices and Modern Methods, pp. 87–104. Springer, Cham (2024)

[6] Tchupo, D.E., Macht, G.A.: Comparing fuzzy cognitive maps: Methods and their applications in team communication. International Journal of Industrial Ergonomics **92**, 103344 (2022)

[7] Amirkhani, A., Papageorgiou, E.I., Mohseni, A., Mosavi, M.R.: A review of fuzzy cognitive maps in medicine: Taxonomy, methods, and applications. Computer methods and programs in biomedicine **142**, 129–145 (2017)

[8] Apostolopoulos, I.D., Papandrianos, N.I., Papathanasiou, N.D., Papageorgiou, E.I.: Fuzzy cognitive map applications in medicine over the last two decades: A review study. Bioengineering **11**(2), 139 (2024)

[9] Cole, J.R., Persichitte, K.A.: Fuzzy cognitive mapping: Applications in education. International journal of intelligent systems **15**(1), 1–25 (2000)

[10] Papageorgiou, E.I., Salmeron, J.L.: A review of fuzzy cognitive maps research during the last decade. IEEE transactions on fuzzy systems **21**(1), 66–79 (2012)

[11] Mourhir, A.: Scoping review of the potentials of fuzzy cognitive maps as a modeling approach for integrated environmental assessment and management. Environmental Modelling & Software **135**, 104891 (2021)

[12] Papageorgiou, E., Kontogianni, A.: Using fuzzy cognitive mapping in environmental decision making and management: a methodological primer and an application. International perspectives on global environmental change, 427–450 (2012)

[13] Papageorgiou, E.I.: Fuzzy Cognitive Maps for Applied Sciences and Engineering: from Fundamentals to Extensions and Learning Algorithms vol. 54. Springer,

Heidelberg (2013)

[14] Giabbanelli, P.J., Rice, K.L., Galgoczy, M.C., Nataraj, N., Brown, M.M., Harper, C.R., Nguyen, M.D., Foy, R.: Pathways to suicide or collections of vicious cycles? understanding the complexity of suicide through causal mapping. Social network analysis and mining **12**(1), 60 (2022)

[15] Gray, S.A., Zanre, E., Gray, S.R.: Fuzzy cognitive maps as representations of mental models and group beliefs. In: Fuzzy Cognitive Maps for Applied Sciences and Engineering: From Fundamentals to Extensions and Learning Algorithms, pp. 29–48. Springer, Heidelberg (2013)

[16] Giles, B.G., Findlay, C.S., Haas, G., LaFrance, B., Laughing, W., Pembleton, S.: Integrating conventional science and aboriginal perspectives on diabetes using fuzzy cognitive maps. Social Science & Medicine **64**(3), 562–576 (2007)

[17] Özesmi, U., Özesmi, S.: A participatory approach to ecosystem conservation: fuzzy cognitive maps and stakeholder group analysis in uluabat lake, turkey. Environmental management **31**, 0518–0531 (2003)

[18] Papageorgiou, E.I.: Learning algorithms for fuzzy cognitive maps—a review study. IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews) **42**(2), 150–163 (2011)

[19] Felix, G., Nápoles, G., Falcon, R., Froelich, W., Vanhoof, K., Bello, R.: A review on methods and software for fuzzy cognitive maps. Artificial intelligence review **52**, 1707–1737 (2019)

[20] Stach, W., Pedrycz, W., Kurgan, L.A.: Learning of fuzzy cognitive maps using density estimate. IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics) **42**(3), 900–912 (2012)

[21] Nápoles, G., Ranković, N., Salgueiro, Y.: On the interpretability of fuzzy cognitive maps. Knowledge-Based Systems **281**, 111078 (2023)

[22] Nápoles, G., Salmeron, J.L., Froelich, W., Falcon, R., Leon Espinosa, M., Vanhoenshoven, F., Bello, R., Vanhoof, K.: Fuzzy cognitive modeling: Theoretical and practical considerations. In: Intelligent Decision Technologies 2019: Proceedings of the 11th KES International Conference on Intelligent Decision Technologies (KES-IDT 2019), Volume 1, pp. 77–87 (2020). Springer

[23] Chi, Y., Liu, J.: Learning of fuzzy cognitive maps with varying densities using a multiobjective evolutionary algorithm. IEEE Transactions on Fuzzy Systems **24**(1), 71–81 (2015)

[24] Kolahdoozi, M., Amirkhani, A., Shojaeefard, M.H., Abraham, A.: A novel quantum inspired algorithm for sparse fuzzy cognitive maps learning. Applied

Intelligence **49**, 3652–3667 (2019)

[25] Ding, F., Luo, C.: Structured sparsity learning for large-scale fuzzy cognitive maps. Engineering Applications of Artificial Intelligence **105**, 104444 (2021)

[26] Stylios, C.D., Groumpos, P.P.: A soft computing approach for modelling the supervisor of manufacturing systems. Journal of Intelligent and Robotic Systems **26**(3), 389–403 (1999)

[27] Dickerson, J.A., Kosko, B.: Virtual worlds as fuzzy cognitive maps. Presence: Teleoperators & Virtual Environments **3**(2), 173–189 (1994)

[28] Huerga, A.V.: A balanced differential learning algorithm in fuzzy cognitive maps. In: Proceedings of the 16th International Workshop on Qualitative Reasoning, vol. 2002 (2002)

[29] Papageorgiou, E., Stylios, C.D., Groumpos, P.P.: Active hebbian learning algorithm to train fuzzy cognitive maps. International journal of approximate reasoning **37**(3), 219–249 (2004)

[30] Papageorgiou, E.I., Groumpos, P.P.: A weight adaptation method for fuzzy cognitive map learning. Soft Computing **9**, 846–857 (2005)

[31] Mateou, N.H., Moiseos, M., Andreou, A.S.: Multi-objective evolutionary fuzzy cognitive maps for decision support. In: 2005 IEEE Congress on Evolutionary Computation, vol. 1, pp. 824–830 (2005). IEEE

[32] Stach, W., Kurgan, L., Pedrycz, W., Reformat, M.: Learning fuzzy cognitive maps with required precision using genetic algorithm approach. Electronics Letters **40**(24), 1519–1520 (2004)

[33] Stach, W., Kurgan, L., Pedrycz, W., Reformat, M.: Genetic learning of fuzzy cognitive maps. Fuzzy sets and systems **153**(3), 371–401 (2005)

[34] Koulouriotis, D.E., Diakoulakis, I.E., Emiris, D.M.: Learning fuzzy cognitive maps using evolution strategies: a novel schema for modeling and simulating high-level behavior. In: Proceedings of the 2001 Congress on Evolutionary Computation (IEEE Cat. No. 01TH8546), vol. 1, pp. 364–371 (2001). IEEE

[35] Parsopoulos, K.E., Papageorgiou, E.I., Groumpos, P.P., Vrahatis, M.N.: A first study of fuzzy cognitive maps learning using particle swarm optimization. In: The Congress on Evolutionary Computation (CEC)., vol. 2, pp. 1440–1447 (2003). IEEE

[36] Papageorgiou, E., Groumpos, P.: Optimization of fuzzy cognitive map model in clinical radiotherapy through differential evolution algorithm. International Journal of Biomedical Soft Computing and Human Sciences: the official journal

of the Biomedical Fuzzy Systems Association **9**(2), 25–31 (2003)

[37] Papageorgiou, E.I., Salmeron, J.L.: Methods and algorithms for fuzzy cognitive map-based modeling. In: Fuzzy Cognitive Maps for Applied Sciences and Engineering: from Fundamentals to Extensions and Learning Algorithms, pp. 1–28. Springer, Heidelberg (2013)

[38] Chen, Y., Mazlack, L.J., Lu, L.J.: Inferring fuzzy cognitive map models for gene regulatory networks from gene expression data. In: 2012 IEEE International Conference on Bioinformatics and Biomedicine, pp. 1–4 (2012). IEEE

[39] Chen, Y., Mazlack, L.J., Minai, A.A., Lu, L.J.: Inferring causal networks using fuzzy cognitive maps and evolutionary algorithms with application to gene regulatory network reconstruction. Applied Soft Computing **37**, 667–679 (2015)

[40] Papageorgiou, E.I., Groumpos, P.P.: A new hybrid method using evolutionary algorithms to train fuzzy cognitive maps. Applied Soft Computing **5**(4), 409–431 (2005)

[41] Zhu, Y., Zhang, W.: An integrated framework for learning fuzzy cognitive map using rcga and nhl algorithm. In: 2008 4th International Conference on Wireless Communications, Networking and Mobile Computing, pp. 1–5 (2008). IEEE

[42] Ren, Z.: Learning fuzzy cognitive maps by a hybrid method using nonlinear hebbian learning and extended great deluge algorithm. In: Midwest Artificial Intelligence and Cognitive Science Conference, p. 159 (2012)

[43] Mitchell, M.: An Introduction to Genetic Algorithms. MIT press, Cambridge, Massachusetts (1998)

[44] Dianati, M., Song, I., Treiber, M.: An introduction to genetic algorithms and evolution strategies. Technical report, Technical report, University of Waterloo, Ontario, Canada (2002)

[45] Bernard, D., Cussat-Blanc, S., Giabbanelli, P.: Fast generation of heterogeneous mental models from longitudinal data by combining genetic algorithms and fuzzy cognitive maps (2023)

[46] Bernard, D., Giabbanelli, P.J.: Creating fcm models from quantitative data with evolutionary algorithms. In: Fuzzy Cognitive Maps: Best Practices and Modern Methods, pp. 121–140. Springer, Cham (2024)

[47] Coello, C.A.C.: Theoretical and numerical constraint-handling techniques used with evolutionary algorithms: a survey of the state of the art. Computer methods in applied mechanics and engineering **191**(11-12), 1245–1287 (2002)

[48] Fortin, F.-A., De Rainville, F.-M., Gardner, M.-A., Parizeau, M., Gagné,

C.: DEAP: Evolutionary algorithms made easy. Journal of Machine Learning Research **13**, 2171–2175 (2012)

[49] Hansen, N., Ostermeier, A.: Completely derandomized self-adaptation in evolution strategies. Evolutionary computation **9**(2), 159–195 (2001)

[50] Biedrzycki, R.: Handling bound constraints in cma-es: An experimental study. Swarm and Evolutionary Computation **52**, 100627 (2020)

[51] Dhuliawala, S., Komeili, M., Xu, J., Raileanu, R., Li, X., Celikyilmaz, A., Weston, J.: Chain-of-verification reduces hallucination in large language models. arXiv preprint arXiv:2309.11495 (2023)

[52] Coello, C.A.C.: Theoretical and numerical constraint-handling techniques used with evolutionary algorithms: a survey of the state of the art. Computer methods in applied mechanics and engineering **191**(11-12), 1245–1287 (2002)

[53] Coello, C.A.C.: Constraint-handling techniques used with evolutionary algorithms. In: Proceedings of the Genetic and Evolutionary Computation Conference Companion, pp. 1310–1333 (2022)

[54] Gray, S.A., Gray, S., Cox, L.J., *et al.*: Mental modeler: a fuzzy-logic cognitive mapping modeling tool for adaptive environmental management. In: 2013 46th Hawaii Int. Conf. on System Sciences, pp. 965–973 (2013). IEEE

[55] Parreño, L., Pablo-Martí, F.: Fuzzy cognitive maps for municipal governance improvement. PLOS ONE **19**(2), 1–28 (2024)

[56] Springvloet, L., Lechner, L., Vries, H., Candel, M.J., Oenema, A.: Short- and medium-term efficacy of a web-based computer-tailored nutrition education intervention for adults including cognitive and environmental feedback: Randomized controlled trial. J Med Internet Res **17**(1), 23 (2015)

[57] Robinson, S.: Experimentation: Obtaining Accurate Results. Simulation: the practice of model development and use. Bloomsbury Publishing, London (2014)

[58] Ross, A., Willson, V.L.: One-Way Anova. Basic and Advanced Statistical Tests: Writing Results Sections and Creating Tables and Figures, pp. 21–24. SensePublishers, Rotterdam (2017)

[59] Nápoles, G., Jastrzębska, A., Mosquera, C., Vanhoof, K., Homenda, W.: Deterministic learning of hybrid fuzzy cognitive maps and network reduction approaches. Neural Networks **124**, 258–268 (2020)

[60] Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, H.: Retrieval-augmented generation for large language models: A survey. arXiv preprint arXiv:2312.10997 (2023)